SPINE · NEET-SS / FET / INI-SS

Spinopelvic alignment

1. Why sagittal alignment matters

- **Sagittal malalignment correlates more strongly with pain and disability than coronal deformity** in adult spinal deformity (Glassman, Schwab).
- The spine and pelvis work as a chain to keep the head over the pelvis with minimal muscular effort (**"cone of economy"**, Dubousset). When alignment fails, the body **compensates**: thoracic hypokyphosis, lumbar hyperextension, **pelvic retroversion (increased pelvic tilt)**, **hip extension, knee flexion**, ankle dorsiflexion. Compensation costs energy and causes pain and fatigue.

2. Pelvic parameters (lateral standing radiograph including the femoral heads)

Parameter	Definition	Normal	Notes
Pelvic incidence (PI)	Angle between the line perpendicular to the **S1 endplate at its midpoint** and the line from that midpoint to the **centre of the femoral heads (hip axis)**	About **50-55 degrees** (range about 35-85)	**Fixed morphological parameter** after skeletal maturity (does not change with position); **determines the lumbar lordosis a person needs**
Pelvic tilt (PT)	Angle between the vertical and the line from the hip axis to the S1 endplate midpoint	**Under about 20 degrees** (about 10-20)	**Positional**; **increases (pelvic retroversion) to compensate for sagittal imbalance**

Parameter	Definition	Normal	Notes
Sacral slope (SS)	Angle between the S1 endplate and the horizontal	About 35-45 degrees	Positional; decreases with retroversion

- **Key equation: PI = PT + SS** (geometric).
- **High PI** (large pelvis): needs **more lumbar lordosis**; associated with **isthmic spondylolisthesis** (shear at L5-S1), and with progression of high-grade slips.
- **Low PI**: flatter spine, less lordosis; associated with **Scheuermann**-like flat types and disc degeneration/herniation (Roussouly type 1-2).

3. Spinal parameters

- **Lumbar lordosis (LL):** Cobb angle **L1 to S1** (about 40-60 degrees; **two-thirds of lordosis between L4 and S1**).
- **Thoracic kyphosis (TK):** T4 (or T5) to T12, about **20-50 degrees**.
- **Sagittal vertical axis (SVA):** horizontal distance from the **C7 plumb line** to the **posterosuperior corner of S1**: **normal under about 4-5 cm** (positive = C7 anterior to S1). **Over 5 cm = positive sagittal imbalance** (disability rises steeply).
- **T1 pelvic angle (TPA):** angle between the line from the femoral heads to T1 centroid and the line from the femoral heads to the S1 endplate midpoint; combines SVA and PT and is **independent of posture compensation** (target **under about 14-20 degrees**).
- **Cervical parameters:** C2-C7 lordosis, **C2-C7 SVA (cSVA; under about 4 cm)**, **T1 slope minus cervical lordosis (T1S-CL under about 20 degrees)**, chin-brow vertical angle (CBVA: horizontal gaze; important in ankylosing spondylitis osteotomy planning).
- **Global alignment and proportion (GAP) score** (Yilgor): relative (PI-adjusted) parameters predict mechanical complications after deformity surgery better than absolute targets.

4. Alignment targets (SRS-Schwab)

- **PI-LL mismatch under about 10 degrees** (LL roughly equal to PI ± 10 degrees).
- **PT under about 20 degrees.**
- **SVA under about 5 cm** (under 4 cm in the SRS-Schwab modifier).
- **Age-adjusted targets** (Lafage): older patients tolerate (and need) more positive alignment; **over-correction in the elderly increases proximal junctional kyphosis (PJK)**.

SRS-Schwab sagittal modifier	0	+	++
PI-LL	Under 10 degrees	10-20	Over 20
PT	Under 20 degrees	20-30	Over 30
SVA	Under 4 cm	4-9.5 cm	Over 9.5 cm

- Coronal curve types: **T** (thoracic only, lumbar under 30), **L** (TL/lumbar only), **D** (double), **N** (no major coronal curve).

5. Roussouly classification of normal sagittal shapes

Type	Sacral slope	Lordosis	Notes
1	**Under 35 degrees**	Short, low lordosis; long thoracolumbar kyphosis	Disc degeneration at the TL junction, posterior element impingement at L5-S1
2	**Under 35 degrees**	Flat, long hypolordosis ("flat back")	**Disc herniation, early disc degeneration**
3	**35-45 degrees**	**Balanced, harmonious** (commonest)	

Type	Sacral slope	Lordosis	Notes
4	**Over 45 degrees**	Long, high lordosis	Facet arthritis, **spondylolysis**, posterior element overload
3 anteverted			Variant with low PI and anteverted pelvis

6. Compensation and decompensation

- **Stages:** (1) **thoracic hypokyphosis**, (2) **hyperextension of mobile lumbar segments** (can cause retrolisthesis, facet overload), (3) **pelvic retroversion (PT rises)**: the most important compensatory mechanism, limited by **hip extension reserve** (hip flexion contractures limit compensation), (4) **knee flexion** and ankle dorsiflexion (energy-expensive, "crouched" posture). When exhausted: **SVA becomes positive**, forward stoop, fatigue, inability to stand upright.
- **Flatback syndrome:** loss of lumbar lordosis after **distraction instrumentation (Harrington rods)** into the lumbar spine, after fusions in kyphosis, or degenerative; forward-leaning posture, knee flexion, back pain.
- **Hip-spine syndrome:** coexisting hip and spine pathology; **spinopelvic mobility affects THA**: a **stiff spine (fused or degenerative) with little pelvic tilt change between standing and sitting** raises the risk of **THA dislocation and impingement** (adjust cup anteversion/inclination; consider dual-mobility). **Sitting: pelvis normally tilts posteriorly (retroverts) about 20 degrees, increasing functional cup anteversion.** "Stuck sitting/stuck standing" pelvises.
- **Proximal junctional kyphosis (PJK): proximal junctional angle over 10 degrees and at least 10 degrees greater than pre-**

operatively (Glattes). Risk factors: older age, osteoporosis, **large corrections**, over-correction relative to age-adjusted targets, fusion to the sacrum, **UIV at the thoracolumbar junction**, posterior ligament disruption, high BMI. Proximal junctional failure (fracture, instability, neurological deficit) needs revision. Prevention: teriparatide, tethers, transitional rods, cement augmentation at UIV, soft landing.

7. Corrective osteotomies

Osteotomy	Schwab grade	Correction	Requirement
Smith-Petersen / Ponte (posterior column)	1-2	**About 10 degrees per level** (about 1 degree per mm of resection)	**Mobile anterior column (open disc)**; opening wedge anteriorly
Pedicle subtraction osteotomy (PSO)	3	**About 30-35 degrees** at one level	Fixed (fused) anterior column; three-column **closing wedge** through the vertebra (usually **L3**, below the conus)
Extended PSO / bone-disc-bone	4	About 35-45 degrees	Includes the disc
Vertebral column resection (VCR)	5-6	**Over 40 degrees**, multiplanar	Rigid, severe, angular deformity (congenital kyphoscoliosis, post-TB) ; **highest neurological risk**

- **Ankylosing spondylitis chin-on-chest deformity:** lumbar PSO (L3) for global kyphosis; **cervicothoracic extension osteotomy (C7-T1)** for cervical kyphosis (C7 has a larger canal and the vertebral artery enters at C6); correct to restore horizontal gaze (**CBVA about 0-10 degrees**).

8. Clinical relevance summary

- **Degenerative spondylolisthesis/stenosis surgery:** restore segmental lordosis (especially L4-S1) during fusion; **interbody devices (ALIF/LLIF) give more lordosis than TLIF/PLIF**.
- **Fusing in kyphosis** leads to **adjacent segment disease** and flatback.
- **Lumbar fusion and THA:** fused lumbar spine increases THA dislocation risk; consider dual-mobility cups.
- **Children's isthmic spondylolisthesis:** high PI and high slip angle predict progression; reduction of high-grade slips (restores alignment but **L5 nerve root palsy** risk).

> Exam pearl: **PI = PT + SS. PI is fixed (morphological); PT and SS are positional. High PI = more lordosis needed and associated with isthmic spondylolisthesis.**

> Exam pearl: **Targets: PI-LL under 10 degrees, PT under 20, SVA under 5 cm (TPA under 20). Pelvic retroversion (high PT) is the key compensation; knee flexion is the last. Sagittal imbalance correlates best with disability.**

> Exam pearl: **Ponte/SPO about 10 degrees per level (needs a mobile disc), PSO about 30-35 degrees at one level (usually L3), VCR over 40 degrees (highest neuro risk). Stiff spine + THA = higher dislocation risk.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.